Team Project & Task Management System

Backend changes required for reliable task loading and signup approval workflow

Prepared from the supplied Spring WebFlux backend source (API Gateway, Task Service, Project Service, User Service and Auth Service).

1. Confirmed cause of the 400 task errors

The Task Service pagination utility explicitly accepts page sizes from 1 to 100. The frontend was sending size=200 to both /tasks/my and /tasks/project/{projectId}; this produces HTTP 400 “size must be between 1 and 100”. The frontend has now been corrected to send size=100. No backend change is needed for this particular 400 error.

The current UI still has to request tasks project-by-project for administrators/project managers. That is inherently more network calls than a single aggregated endpoint. The recommended endpoint below is optional but strongly recommended for performance.

2. Recommended task aggregation endpoint

Add a role-aware endpoint in Task Service and route it through API Gateway:

```
GET /tasks/visible?page=0&size=100&sort=dueDate,asc
```

Expected behavior:

• TEAM_MEMBER: return only tasks assigned to the authenticated user.

• PROJECT_MANAGER: return tasks in projects the manager is authorized to access.

• ADMIN: return tasks the administrator is authorized to view according to the project-access policy.

• Return a standard PageResponse with content, page, size, totalElements and totalPages. Enforce size <= 100 server-side.

• Use the authenticated principal from the existing gateway/JWT context; do not accept userId or role as trusted client query parameters. Apply existing project/task authorization checks.

3. Signup request / admin approve-reject workflow

The supplied UserRouter currently has /users/me, /users/directory and admin user-management routes, but no public registration-request route and no admin request-review routes. The frontend cannot complete approval/rejection until these backend endpoints exist.

Method and path	Access	Purpose
POST /users/registration-requests	Public (gateway allowlist)	Submit name, email, password, optional department/phone; create PENDING request only.
GET /admin/registration-requests?status=PENDING&page;=0&size;=100	ADMIN only	List pending requests.
PATCH /admin/registration-requests/{id}/approve	ADMIN only	Approve request, create/activate user and assign an allowed role.
PATCH /admin/registration-requests/{id}/reject	ADMIN only	Reject request and store optional reason.

Suggested persistence model

```
registration_requests
- id (primary key)
- name
```

```
- email (normalized; unique while pending)
- password_hash (never store plaintext)
- department, phone (nullable)
- status: PENDING | APPROVED | REJECTED
- requested_at, reviewed_at
- reviewed_by (admin user id, nullable)
- rejection_reason (nullable)
```

Security and consistency requirements

• Hash the password with the same approved PasswordEncoder used by User Service; never log or return it.

• Validate input server-side, normalize email, rate-limit public requests, and avoid duplicate pending requests.

• Approval must be transactional/idempotent: only PENDING requests can be approved; create user once, enforce unique email, and record reviewer/time. Do not allow the requester to choose ADMIN.

• Reject and approve routes must be protected by role-based authorization in both gateway and service. Only the public POST request endpoint should be unauthenticated.

• If browser uses HttpOnly cookies, preserve existing cookie/CORS/CSRF policy. Do not expose access tokens in request records.

4. Why loading can still take time

Fixing size=200 removes the confirmed 400 errors. For ADMIN/PROJECT_MANAGER, the frontend still fetches a project list and then one task page per project. The /tasks/visible endpoint would reduce that to one task request. For more than 100 tasks, implement page iteration or load additional pages using totalPages; do not send size greater than 100.

5. Files reviewed

TaskRouter / TaskHandler / Params; ProjectRouter / ProjectHandler; UserRouter / UserHandler; Auth Service security and gateway route/security configuration in the supplied source archive. The existing public backend does not currently expose the registration request workflow.